**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# LESSON 122 · Validation and Authentication

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. After correcting an assessment answer, why attempt a new question?

Q2. After one ascending bubble-sort pass on [7, 3, 6, 2], what is the list?
A. [2, 3, 6, 7]
B. [3, 6, 2, 7]
C. [7, 3, 6, 2]
D. [2, 6, 3, 7]

Q3. “An IF statement must execute every possible branch regardless of its condition.” Correct this. Explain why.

Q4. A circuit has X = A AND B, followed by Q = X OR C. For input rows 000, 011, 110, 111, give the pairs (X,Q). Why is a column for X useful?

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# I Do · Read new knowledge

## Key vocabulary

Validation — Checks whether input meets a defined rule, not whether it is factually true.

Range check — Rejects a number outside stated minimum and maximum boundaries.

Presence and type — Detects missing required input or wrong type/format.

Authentication — Checks a user's claimed identity, which differs from validating an entered value.

## New learning

Validation: Checks whether input meets a defined rule, not whether it is factually true.

Range check: Rejects a number outside stated minimum and maximum boundaries.

Presence and type: Detects missing required input or wrong type/format.

Authentication: Checks a user's claimed identity, which differs from validating an entered value.

## Worked example

Design a range check for an integer mark from 0 to 100 inclusive.

1. Input the candidate mark.

2. Test mark >= 0 AND mark <= 100.

3. Accept only when both conditions are true.

4. Otherwise display an error and ask for a new input.

# We Do · Practise with guidance

### Guided 1

A form accepts ages from 13 to 18. Write an acceptance condition.

Worked method: age >= 13 AND age <= 18.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Code First · Predict → Write → Test → Improve

Finish the inclusive range check for one IPv4 octet.

```
octet = 255
if ... then
    print("Valid range")
else
    print("Invalid")
endif
```

write

### Question 1

Explain validation versus verification. [2 marks]

### Question 2

Give a suitable boundary value when valid ages are 13–18. [2 marks]

### Question 3

State one difference between authentication and validation. [2 marks]

# Exit Ticket · Answer before checking

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Validation proves a pupil's entered date is factually correct.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. To check that the method can be applied independently rather than copied.

Q2. B — [3, 6, 2, 7]

Q3. It chooses a branch using a condition. Selection can be nested.

Q4. (a) (0,0), (0,1), (1,1), (1,1). X shows the first gate's output. This helps work out Q and find errors.

### I Do / We Do pages · Guided practice

Q1. age >= 13 AND age <= 18.

### You Do pages · Code First

```
octet = 255
if octet >= 0 AND octet <= 255 then
    print("Valid range")
else
    print("Invalid")
endif
```

Expected result: Valid range

### You Do pages · Independent questions

Q1. Validation checks a rule or format; verification checks data matches the original/source input.

Q2. 13 or 18, plus an adjacent invalid such as 12 or 19 for a test set.

Q3. Authentication confirms an asserted identity; validation checks input against constraints.

### Exit Ticket · Model answer

False

# Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

## 1. Copy the whole original question:

## 2. Write a complete new answer in GREEN PEN. Show working if needed: